

A COMMON-SOURCE AND SCALE-RIGIDIFIED RECONSTRUCTION OF THE IUT_{–abc} BRIDGE A PROOF CANDIDATE AFTER THE 2026 FORMALIZATION LANDSCAPE, WITH ORDINARY-MATHEMATICS RECONSTRUCTION AND LEAN VERIFICATION OF THE ABSTRACT CORRIDOR

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ABSTRACT. This paper develops a strengthened proof candidate for the *abc* conjecture by rewriting the decisive passage from IUT III, Theorem 3.11 to Corollary 3.12 in ordinary categorical, Picard-theoretic, and measure-theoretic language. Two synchronization problems are isolated and treated.

First, the horizontal link is parameterized by one common source isomorphism. The same source parameter induces the structured horizontal map, its unit and value-group portions, the associated coric/procession transport, and the Kummer comparison maps. This replaces post hoc assembly of unrelated existential witnesses by a single coherent tuple.

Second, the comparison of global realified Frobenioids is initially defined only up to a positive real degree-scaling orbit. Packet-normalized log-volume rigidifies this orbit: multiplication by a rational prime p changes the signed local log-volume by $-\log p$ on both sides. If a comparison scaled arithmetic degree by $\lambda > 0$, compatibility with this normalized response would give $-\lambda \log p = -\log p$, hence $\lambda = 1$. Thus the pilot-preserving comparison has a scale-one representative.

Choosing the ordinary Kummer branch of Ind3 and globalizing the resulting local fractional ideals produces a candidate actual theta-pilot image W satisfying

$$W \in \mathcal{U}_\Theta, \quad \mu_{\text{proc}}(W) = -|\log q|.$$

Monotonicity then gives

$$-|\log q| \leq -|\log \Theta|,$$

which is the missing numerical inequality in IUT III, Corollary 3.12. The remaining scalar and arithmetic-geometric chain is the one stated in IUT IV and specializes to the logarithmic *abc* inequality on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$.

The manuscript also compares this route with the current Lean formalization programmes, Arithmetic Teichmüller Spaces, analytic exceptional-set bounds, and the function-field Mason–Stothers/Vojta analogues. Lean 4 verifies the abstract common-source selection, single-link conjugation, scale-rigidity calculation, principal-degree invariance, ordered-volume bridge, and scalar implication $C_\Theta \geq -1$. It does not yet encode the complete definitions of Hodge theaters, structured prime-strips, or the concrete Ind1–Ind3 relation. The result is therefore a closed source-grounded proof candidate for expert scrutiny, not a settled machine-certified resolution.

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Verification status

The mathematical corridor after a coherent scale-one source witness is completely explicit and its abstract form is Lean-checked. The remaining verification target is source-semantic: the functorial maps in IUT II, Corollary 4.10 and IUT III, Theorem 3.11 must instantiate the common-source and scale-one hypotheses exactly as used below. Current public formalization projects explicitly retain this construction behind an interface or input region. Consequently, this paper should be read as a candidate closure and a finite verification programme, not as a claim that the international mathematical community or Lean’s kernel has already certified the full IUT-to-*abc* chain.

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1. INTRODUCTION

The *abc* conjecture is a statement about the tension between additive structure and multiplicative complexity. For pairwise coprime nonzero integers a, b, c with $a + b = c$, set

$$H(a, b, c) = \log \max\{|a|, |b|, |c|\}, \quad R(a, b, c) = \sum_{p|abc} \log p = \log \text{rad}(abc).$$

The conjecture asserts that, for every $\varepsilon > 0$, there is a constant C_ε such that

$$(1) \quad H(a, b, c) \leq (1 + \varepsilon)R(a, b, c) + C_\varepsilon.$$

Mochizuki’s four IUT papers construct a system of arithmetic objects intended to separate, transport, and then compare the additive and multiplicative structures underlying a number field equipped with an elliptic curve. The first paper constructs Hodge theaters and non-scheme-theoretic theta-links; the second constructs theta evaluation and the Gaussian family $q_v^{j^2}$; the third constructs log-links, log-shells, multiradial representations, and the crucial log-volume estimate; the fourth converts that estimate into Diophantine inequalities. Mochizuki describes the division of labor in this manner: the first three papers primarily construct new objects, the final part of the third establishes the decisive properties, and the fourth combines those properties with explicit computations to obtain Diophantine consequences [1, 2, 3, 4].

The disputed numerical passage is the implication

$$(2) \quad -|\log q| \leq -|\log \Theta|.$$

Once (2) is known, IUT III, Corollary 3.12 follows immediately: if

$$-|\log \Theta| \leq C_\Theta |\log q|, \quad |\log q| > 0,$$

then

$$-|\log q| \leq C_\Theta |\log q|,$$

so $C_\Theta \geq -1$. IUT IV inserts this coefficient bound into explicit local and global estimates and derives a Vojta-type inequality for hyperbolic curves, from which *abc* follows as the case $(X, D) = (\mathbb{P}^1, \{0, 1, \infty\})$.

The purpose of this paper is to express the bridge (2) in standard mathematics. The main structural idea is that a strict cross-column comparison in F_{mod} is the wrong target: F_{mod} remembers additive local fractional ideals and is only upper semi-compatible across log-links. The exact comparison should first be performed in F_{MOD} , which depends only on global multiplicative data and localization homomorphisms. After the target column has been reached, one applies $F_{\text{MOD}} \simeq F_{\text{mod}}$ once, in that single column. This is the *same-column descent principle*.

The second key point is categorical. Mochizuki’s “full poly-isomorphism” means the full collection of isomorphisms between the relevant structured objects. If K_0 , K_1 , and P are the Kummer and column-permutation equivalences, the conjugate

$$H_* = K_1^{-1} P K_0$$

is automatically an allowed horizontal representative. Its counit gives

$$K_1 H_* \cong P K_0.$$

The proof then evaluates this natural isomorphism at the theta-pilot, descends once to a target-column fractional-ideal object, and uses the product formula to identify its degree with the degree of the native q -pilot.

For coherence it is preferable to reverse this choice. Choose one horizontal representative h first and set

$$P_h = K_1 h K_0^{-1}.$$

Then the commuting triangle is definitional up to the unit/counit of the Kummer equivalences, and the same h supplies the unit and value-group portions. The fullness of the coric étale-picture transport is used only to show that P_h is allowed. This “single-link-first” formulation directly addresses the objection that independently selected component witnesses need not assemble.

2. THE 2026 RESEARCH LANDSCAPE AND ROUTE SELECTION

The most relevant recent developments sharpen the verification boundary rather than remove it.

- (i) The public LANA/IUT Lean repository explicitly states that it does not verify IUT. Its Corollary 3.12 strand defines the target inequality and a large-volume container, but the theta-pilot region is input data and no multiradial algorithm constructing it is proved.
- (ii) The July 2026 LANA interim report states that no final judgment has been made on the correctness of IUT; the main achievement is the conversion of vague controversies into diagrams, types, code, black boxes, and explicit questions.
- (iii) Joshi's Arithmetic Teichmüller Spaces programme claims a proof of Corollary 3.12 and a Rosetta-stone comparison with IUT. This is a valuable independent route, but its identification of the constructed theta locus with the IUT possible-image locus remains a separate theorem to be checked.
- (iv) Analytic number theory now gives a power-saving exceptional-set estimate: Browning–Lichtman–Teräväinen prove $O(X^{33/50})$, and Li improves this to $O(X^{56/85+\varepsilon})$. Such estimates do not imply finiteness of an arbitrarily sparse exceptional set.
- (v) Mason–Stothers, the function-field *abc* theorem, has been formalized in Lean 4, and Vojta-type *abc* statements are known for several function-field toric settings. These results confirm the geometric architecture but do not supply a nonzero global derivation on $\mathbb{Z}$ or a number-field truncated counting theorem.

The route pursued below is therefore the one closest to the final target: construct, from a single source parameter in IUT II–III, one scale-rigidified possible image of the theta-pilot whose reading is the native q -pilot reading.

3. AN ORDINARY-MATHEMATICS DICTIONARY FOR IUT

This section translates the IUT vocabulary used later into standard mathematical objects. It is not a substitute for the detailed definitions in [1, 2, 3]; it identifies the conventional role played by each object in the proof.

Definition 3.1 (Hodge theater). A Hodge theater is a structured copy of the arithmetic geometry attached to the initial theta-data. For the present argument it may be viewed as a package containing:

- (i) local and global Frobenioids encoding divisors and line bundles;
- (ii) arithmetic fundamental groups and their localization data;
- (iii) label sets carrying additive and multiplicative symmetries;
- (iv) Kummer comparison maps between Frobenius-like and étale-like descriptions.

Distinct Hodge theaters are abstractly isomorphic but are treated as carrying distinct arithmetic holomorphic structures.

Definition 3.2 (Frobenioid). For this paper, a Frobenioid is used as a categorical encoding of arithmetic line bundles, divisor monoids, rational functions, and Frobenius tensor powers. The two global realizations relevant here are:

- (i) F_{MOD} , the rational-function torsor realization, which is biased toward multiplicative data;
- (ii) F_{mod} , the local fractional-ideal realization, which is biased toward additive integral structures and permits explicit degree and volume calculations.

The source gives a natural equivalence $F_{\text{MOD}} \simeq F_{\text{mod}}$ inducing the identity on the rational-function monoid.

Definition 3.3 (Theta-link and log-link). A theta-link is a non-ring-theoretic identification of selected multiplicative/Frobenioid data in two distinct Hodge theaters. A log-link is the local passage induced by the p -adic or archimedean logarithm. The theta-link moves horizontally between arithmetic holomorphic structures; the log-link moves vertically within one such structure.

Definition 3.4 (Pilot objects). A q -pilot is the global divisor or line object determined by the local Tate parameters q_v . A theta-pilot is the global object determined by the collection of theta values $q_v^{j^2}$. The relevant horizontal LGP-link sends theta-pilot objects to q -pilot objects.

Definition 3.5 (Multiradial representation). A multiradial representation expresses objects originating in one Hodge theater in a mono-analytic container that can be interpreted from another Hodge theater. The representation is subject to three kinds of indeterminacy:

- (Ind1) permutations of procession labels and automorphisms of the corresponding mono-analytic prime-strips;
- (Ind2) automorphisms of the units-mod-torsion part of the horizontal link;
- (Ind3) upper semi-compatible Kummer transport along the log-link.

The theta-pilot log-volume is the volume of the union of all possible images under these indeterminacies.

Remark 3.6. In conventional language, the decisive task is therefore simple to state: construct one actual region W among the allowed theta-pilot images, prove that its procession-normalized signed log-volume equals the native q -pilot degree, and use monotonicity.

4. THE PROOF ARCHITECTURE

Let $\mathcal{U}_\Theta$ be the union of all theta-pilot possible images in the target mono-analytic container, and let

$$\mu_{\text{proc}}(\mathcal{U}_\Theta) = -|\log \Theta|.$$

Let the native q -pilot region have volume $-|\log q|$.

Theorem 4.1 (Same-column full-poly volume witness). *In the setting of IUT III, Theorem 3.11, for the horizontal arrow from $(0,0)$ to $(1,0)$, there is an actual allowed theta-pilot image W in the target 1-column such that*

$$W \subseteq \mathcal{U}_\Theta, \quad \mu_{\text{proc}}(W) = -|\log q|.$$

Consequently,

$$-|\log q| \leq -|\log \Theta|.$$

The proof occupies [Sections 6 to 10](#). It is useful first to explain two targets that are too strong.

5. WHY A STRICT CROSS-COLUMN SQUARE IS THE WRONG TARGET

5.1. Coarsified categorical equality. Let $\mathbf{hEqCat}$ denote the homotopy 1-category whose objects are categories and whose morphisms are equivalence functors modulo natural isomorphism. If $[F] = [G]$ in $\mathbf{hEqCat}(\mathcal{C}, \mathcal{D})$, then every $X \in \mathcal{C}$ carries an isomorphism $F(X) \cong G(X)$. Every isomorphism-invariant numerical reading therefore has the same value on $F(X)$ and $G(X)$.

Mochizuki's convention is of this coarsified kind: an isomorphism between categories is an equivalence class of equivalences, while a full poly-isomorphism between structured objects is the collection of all isomorphisms between them [[1](#), Section 0]. The argument therefore needs a natural-isomorphism class, not a strict equality of functors.

5.2. The value-group dilation obstruction. The horizontal link preserves the unit portion only up to isomorphism; it does not preserve the divisor/value-group portion. Indeed, the Gaussian map contains the nontrivial dilation

$$q_v \longmapsto (q_v^{j^2})_{1 \leq j \leq \ell^*}.$$

A single fixed degree reading cannot identify q and every q^{j^2} by degree-preserving isomorphisms.

Proposition 5.1 (Fixed-scale obstruction). *Let $Q > 0$. There is no degree-preserving identification that simultaneously reads $\deg(q) = Q$ and identifies every q^{j^2} with q in one fixed degree scale.*

Proof. Taking $j = 2$ gives $\deg(q^4) = 4Q$ by additivity. A degree-preserving identification with q would give $4Q = Q$, hence $Q = 0$, a contradiction. $\square$

Thus the horizontal comparison must be made before fixing the final additive degree realization.

5.3. Why not descend to F_{mod} first? The construction of F_{MOD} uses only the multiplicative group of the global field, its local value groups, their positive cones, and the valuation homomorphisms. The construction of F_{mod} uses local fractional ideals and therefore the local additive structures. IUT III explicitly distinguishes the two: the MOD presentation admits precise log-Kummer comparison, whereas the mod presentation is subject to upper semi-compatibility and substantial distortion [3, Introduction and Example 3.6].

Accordingly, we do not demand a strict cross-column square in F_{mod} . Exact comparison is first completed in F_{MOD} ; then the equivalence

$$N_1 : F_{\text{MOD}}(1, \circ) \xrightarrow{\sim} F_{\text{mod}}(1, \circ)$$

is applied once in the target column.

6. THE FUNCTORIAL CONTENT OF F_{MOD}

Definition 6.1 (Global multiplicative datum). A global multiplicative datum is a tuple

$$\mathfrak{D} = (M, V, \{\Gamma_v, \Gamma_v^+, \beta_v\}_{v \in V})$$

consisting of an abelian group M , an index set V , an abelian group Γ_v with a positive submonoid Γ_v^+ for each v , and a homomorphism $\beta_v : M \rightarrow \Gamma_v$. For a number field one has

$$M = F^\times, \quad \Gamma_v = F_v^\times / \mathcal{O}_{F_v}^\times,$$

with Γ_v^+ the nonnegative valuation cone.

An isomorphism of such data consists of an isomorphism of M , a bijection of V , and compatible isomorphisms of the Γ_v preserving the positive cones and commuting with the β_v . These data form a groupoid **GMD**.

Construction 6.2 (The rational-function torsor category). For $\mathfrak{D} \in \mathbf{GMD}$, define $\mathbf{F}_{\text{MOD}}(\mathfrak{D})$ as follows. An object is an M -torsor T together with a trivialization of every induced Γ_v -torsor $T \times^M \Gamma_v$. An elementary morphism is an M -equivariant torsor isomorphism whose local displacement lies in Γ_v^+ at every v . A general morphism is obtained after a positive Frobenius tensor power. This is the conventional content of the rational-function torsor presentation in IUT III, Example 3.6(i).

Proposition 6.3 (F_{MOD} is functorial). *An isomorphism $\Phi : \mathfrak{D} \rightarrow \mathfrak{D}'$ induces an equivalence*

$$\mathbf{F}_{\text{MOD}}(\Phi) : \mathbf{F}_{\text{MOD}}(\mathfrak{D}) \xrightarrow{\sim} \mathbf{F}_{\text{MOD}}(\mathfrak{D}'),$$

and the construction is compatible with composition up to the canonical natural isomorphism. Hence it defines a functor $\mathbf{GMD} \rightarrow \mathbf{hEqCat}$.

Proof. Extend the structure group of an M -torsor along $M \xrightarrow{\sim} M'$. Compatibility with each β_v transports the local trivializations uniquely. Preservation of the positive cones preserves the integrality condition on elementary morphisms, and Frobenius tensor powers are preserved. The inverse datum produces a quasi-inverse. Associativity of extension of structure group supplies the natural isomorphism required for functoriality. $\square$

Lemma 6.4 (Pilot transport). *If Φ sends the generator data defining a pilot object $T(a)$ to the generator data a' , then*

$$\mathbf{F}_{\text{MOD}}(\Phi)(T(a)) \cong T(a').$$

Proof. The local trivialization of $T(a)$ is displaced from a reference trivialization by a_v . After extension of structure group, that displacement is $\Phi_v(a_v) = a'_{\sigma(v)}$, which is precisely the trivialization defining $T(a')$. $\square$

7. FULL-POLY CONJUGATION AND THE STRUCTURED SQUARE

7.1. The categorical lemma.

Lemma 7.1 (Full-poly conjugation). *Let*

$$K_0 : \mathcal{A}_0 \xrightarrow{\sim} \mathcal{C}_0, \quad K_1 : \mathcal{A}_1 \xrightarrow{\sim} \mathcal{C}_1, \quad P : \mathcal{C}_0 \xrightarrow{\sim} \mathcal{C}_1$$

be equivalences. Suppose the allowed horizontal arrows from $\mathcal{A}_0$ to $\mathcal{A}_1$ form the full poly-isomorphism, namely the collection of all equivalences. Then there is an allowed representative

$$H_* = K_1^{-1} P K_0$$

and a natural isomorphism

$$K_1 H_* \xrightarrow{\sim} P K_0.$$

Proof. The composite H_* is an equivalence, hence belongs to the full collection by definition. The counit $K_1 K_1^{-1} \Rightarrow \text{id}_{\mathcal{C}_1}$, whiskered by $P K_0$, gives

$$K_1 K_1^{-1} P K_0 \xrightarrow{\sim} P K_0.$$

This is the required natural isomorphism. Different choices of quasi-inverse alter only the natural-isomorphism class. $\square$

Remark 7.2 (Necessity of fullness). If the allowed horizontal family is a proper subset of all equivalences, the conjugate $K_1^{-1} P K_0$ need not belong to it. A one-object groupoid with automorphism group $\mathbb{Z}/2\mathbb{Z}$ gives an immediate counterexample: allow only the identity horizontally and choose the conjugate to be the nontrivial automorphism.

7.2. A common source parameter for all components. IUT II, Corollary 4.10(iii), starts with a nonempty set

$$\mathcal{S} = \text{Isom}_F(F_{\text{env}}^{(0)}, F_{\text{cns}}^{(1)})$$

of isomorphisms of underlying F -prime-strips and applies functorial algorithms to obtain structured $F^{\triangleright \times \mu}$ -isomorphisms. The Gaussian version is obtained from the same source map by composition with the evaluation isomorphism. Proposition 3.7 supplies the natural Gaussian-to-LGP comparison. Thus one may choose

$$s \in \mathcal{S}$$

and derive from this one parameter:

$$h_s, \quad h_s^{\times \mu}, \quad h_s^{\triangleright}, \quad d_s, \quad p_s,$$

namely the complete horizontal structured representative, its unit and value-group portions, its underlying D -prime-strip map, and the induced procession transport.

Theorem 7.3 (Common-source synchronization). *Assume the functorial maps and compatibility assertions in IUT II, Corollary 4.10 and IUT III, Theorem 3.11 have their standard pointwise meaning. Then there exists one source parameter $s \in \mathcal{S}$ for which all unit, value-group, splitting-monoid, number-field, localization, and procession maps belong to one structured commuting diagram. The value-group component maps the theta-pilot to a q-pilot.*

Proof. Choose s from the nonempty source set. Every displayed component is obtained from s by a functorial construction or by taking a functorial projection of the resulting structured prime-strip isomorphism. The evaluation isomorphism relates the environmental and Gaussian constructions before the structured link is formed. Theorem 3.11(iii)(a)–(d) states compatibility of the Kummer maps with the horizontal maps on precisely the unit, theta/splitting, number-field, and localization data. Hence these are not unrelated existential witnesses: they

are the images of one s . Finally, Definition 3.8 and Remark 3.8.1 identify the image of a theta-pilot under the value-group part with a q -pilot. $\square$

The theorem is exactly the point that an eventual full formalization must encode: “compatible full poly-isomorphisms” must be realized as a relation with a common parameter, not merely as a list of surjective projections.

7.3. Single-link-first conjugation. The preceding lemma chooses P first. The source proof of Corollary 3.12, however, insists that the unit and value-group data be treated through one single horizontal link. The following equivalent formulation is better adapted to that requirement.

Lemma 7.4 (Single-link-first conjugation). *Let*

$$h : \mathcal{A}_0 \xrightarrow{\sim} \mathcal{A}_1, \quad K_0 : \mathcal{A}_0 \xrightarrow{\sim} \mathcal{C}_0, \quad K_1 : \mathcal{A}_1 \xrightarrow{\sim} \mathcal{C}_1$$

be equivalences. Define

$$P_h := K_1 h K_0^{-1} : \mathcal{C}_0 \xrightarrow{\sim} \mathcal{C}_1.$$

Then

$$P_h K_0 \xRightarrow{\sim} K_1 h.$$

If the allowed transports from $\mathcal{C}_0$ to $\mathcal{C}_1$ form the full poly-isomorphism, then P_h is allowed. If $h(T_\Theta) \cong Q_q$, then

$$P_h K_0(T_\Theta) \cong K_1(Q_q).$$

Proof. The natural isomorphism is obtained by cancelling the adjacent pair $K_0^{-1} K_0$ with the unit of the equivalence K_0 . Fullness includes every equivalence, hence includes P_h . Evaluating the natural isomorphism at T_Θ and using $h(T_\Theta) \cong Q_q$ proves the final assertion. $\square$

This formulation has a useful logical feature: no independent unit witness and value witness are ever selected. They are projections of one fixed structured arrow h .

7.4. Ind2 insulation and coherent assembly. Projection-wise compatibility is insufficient in general. The anti-diagonal relation $\{(1, 0), (0, 1)\}$ has both desired projections but no output $(1, 1)$. The source contains an additional property that rules out this pattern: the Ind2 automorphism group acts on the unit portion, while the cyclotomic/value and number-field portions are insulated from that action.

Lemma 7.5 (Insulated-orbit coherence). *Let a group G act on a set U , and let $v_0 \in V$. Suppose the structured outputs generated by the Ind2 ambiguity are*

$$\mathcal{O} = \{(g \cdot u_0, v_0) : g \in G\} \subseteq U \times V.$$

Then every allowed unit representative is paired with the same value component v_0 . In particular, the anti-diagonal obstruction cannot occur.

Proof. The second coordinate of every displayed element is, by definition, v_0 . $\square$

In IUT III, the source explicitly describes the cyclotomes related to the theta function and number field as insulated from the $\text{Aut}_{F \times \mu}$ indeterminacies acting on the units. Thus the Ind2 orbit has the form of Theorem 7.5. Ind1 acts functorially on the whole procession, hence relabels the complete tuple rather than assembling unrelated projections. Ind3 is relational rather than group-like, but its ordinary Kummer branch is explicitly among the permitted images. These three facts supply the synchronization needed below.

7.5. The four structured arrows. Use the notation

$$\begin{aligned} H_{00}^{\text{MOD}} &: \mathcal{F}_{00}^{\Theta, \text{MOD}} \dashrightarrow \mathcal{F}_{10}^{q, \text{MOD}}, \\ K_{00}^{\text{MOD}} &: \mathcal{F}_{00}^{\Theta, \text{MOD}} \dashrightarrow \mathcal{F}_{0\circ}^{\text{MOD}}, \\ K_{10}^{\text{MOD}} &: \mathcal{F}_{10}^{q, \text{MOD}} \dashrightarrow \mathcal{F}_{1\circ}^{\text{MOD}}, \\ P_{01}^{\text{MOD}} &: \mathcal{F}_{0\circ}^{\text{MOD}} \dashrightarrow \mathcal{F}_{1\circ}^{\text{MOD}}. \end{aligned}$$

Here H_{00} is a representative of the horizontal $\Theta_{\text{LGP}}^{\times\mu}$ -link, K_{n0} are global Kummer equivalences, and P_{01} is induced by an étale-picture permutation. The dashed arrows emphasize that representatives are chosen from full poly-isomorphisms.

The local structured monoid of an $F^{\triangleright\times\mu}$ -prime-strip is assembled from its value-group portion and its units-mod-torsion portion. The global F_{MOD} datum is then determined by the number-field multiplicative group, the local value groups and positive cones, and the localization maps. Consequently, compatibility of all these components for the *same* horizontal representative assembles to a commuting square in **GMD**.

Theorem 7.6 (Single-link structured F_{MOD} square). *Fix one representative*

$$h_{00}^{\text{MOD}} \in \text{FullPolyIso}(\mathcal{F}_{00}^{\Theta, \text{MOD}}, \mathcal{F}_{10}^{q, \text{MOD}})$$

of the horizontal LGP-link. Under the functoriality and compatibility statements of IUT III, Theorem 3.11(i)–(iii), the induced coric transport

$$P_s^{\text{MOD}} := K_{10}^{\text{MOD}} h_{00}^{\text{MOD}} (K_{00}^{\text{MOD}})^{-1}$$

is an allowed étale-picture/procession transport and there is a natural isomorphism

$$(3) \quad P_s^{\text{MOD}} K_{00}^{\text{MOD}} \xrightarrow{\sim} K_{10}^{\text{MOD}} h_{00}^{\text{MOD}}.$$

Moreover,

$$(4) \quad h_{00}^{\text{MOD}}(T_{\Theta}) \cong Q_q.$$

The unit, splitting-monoid, number-field, and localization components in this square arise from the same representative h_{00}^{MOD} .

Proof. The horizontal LGP-link is a full poly-isomorphism of the entire structured $F^{\triangleright\times\mu}$ -prime-strip, not a pair of independently chosen projections. Choose one representative h_{00}^{MOD} . Its underlying map on the coric D -prime-strip/procession determines one étale-picture transport. The multiradial construction of Theorem 3.11(i) is functorial in such procession isomorphisms. The Kummer-detachment algorithms of Theorem 3.11(ii) are functorial, and the four assertions of Theorem 3.11(iii)(a)–(d) state their compatibility, for this horizontal arrow, on the log-shell/unit, splitting-monoid, number-field, and localization components. Hence the component maps are the components of one structured natural isomorphism. Abstractly this is exactly [Theorem 7.4](#), with $P_s^{\text{MOD}} = K_{10}^{\text{MOD}} h_{00}^{\text{MOD}} (K_{00}^{\text{MOD}})^{-1}$.

The source's Ind2-insulation statements show that changing the unit representative does not change the cyclotomic/value or number-field components; thus the anti-diagonal countermodel cannot arise, by [Theorem 7.5](#). Finally, Definition 3.8 and Remark 3.8.1 state that both the LGP and lgp links map theta-pilot objects to q -pilot objects, which gives (4). $\square$

Remark 7.7 (Exact source-facing verification point). The proof uses the standard pointwise meaning of compatibility of structured poly-isomorphisms: after one element h of the full link is chosen, its induced maps on all functorially constructed components belong to one commuting structured diagram. This reading is supported by the source's insistence on one single horizontal link and by the Ind2-insulation statements. A complete Lean formalization must nevertheless encode the actual structured prime-strip category and verify this interpretation from the definitions, rather than taking it as an informal convention.

Evaluating (3) at T_Θ and using (4) gives an isomorphism in the target multiplicative category:

$$(5) \quad P_s^{\text{MOD}} K_{00}^{\text{MOD}}(T_\Theta) \cong K_{10}^{\text{MOD}}(Q_q).$$

8. SCALE RIGIDIFICATION AND SAME-COLUMN DESCENT

8.1. The positive-real orbit ambiguity. The horizontal comparison of global realified Frobenioids is compatible with an $\mathbb{R}_{>0}$ -orbit of isomorphisms. This orbit acts by multiplying arithmetic degrees by a positive scalar. Consequently, an abstract isomorphism of realified Frobenioid objects does not by itself imply equality of their numerical degrees.

Let $\lambda_s > 0$ denote the scale factor of the representative induced by the common source parameter s .

8.2. Packet normalization fixes the scale. For a nonarchimedean packet, the source normalization satisfies

$$\mu_{\log}(\mathcal{O}) = 0, \quad \mu_{\log}(p\mathcal{O}) = -\log p.$$

The analogous archimedean normalization uses multiplication by e .

Lemma 8.1 (Scale rigidity from one normalized response). *Let p be a rational prime and $\lambda > 0$. If a comparison scales arithmetic degree by λ and preserves the packet-normalized response to multiplication by p , then $\lambda = 1$.*

Proof. The source response is $-\log p$. After degree scaling it is $-\lambda \log p$. Preservation gives

$$-\lambda \log p = -\log p.$$

Since $\log p > 0$, division yields $\lambda = 1$. $\square$

Theorem 8.2 (Scale-one source representative). *For the common-source representative of Theorem 7.3, the $\mathbb{R}_{>0}$ scale is equal to one.*

Proof. The Kummer isomorphisms of IUT III, Theorem 3.11(ii)(a), are stated to be compatible with the packet log-volumes. Proposition 3.9(iv) gives compatibility along log-links. Theorem 3.11(iii)(d) and Remark 3.11.2(iv) preserve, and indeed reconstruct, the localization data from the response of packet volumes to multiplication by global number-field elements. In particular, the rational prime p and its normalized response are the same on both sides of the common-source diagram. Apply Theorem 8.1. $\square$

This step is essential: without it the realified comparison would determine only a ray of possible degrees, not the native q reading.

8.3. Descent in the target column. Evaluate the common-source commuting square at T_Θ . In the target multiplicative category one obtains

$$(6) \quad P_s^{\text{MOD}} K_{00}^{\text{MOD}}(T_\Theta) \cong K_{10}^{\text{MOD}}(Q_q).$$

Apply the natural equivalence

$$N_1 : \mathcal{F}_{10}^{\text{MOD}} \xrightarrow{\sim} \mathcal{F}_{10}^{\text{mod}}$$

only after the target column has been reached, and define

$$(7) \quad W_{\text{obj}} = N_1 P_s^{\text{MOD}} K_{00}^{\text{MOD}}(T_\Theta), \quad Q_{\text{obj}} = N_1 K_{10}^{\text{MOD}}(Q_q).$$

By Theorem 8.2, this is a scale-one comparison. The two local fractional-ideal representatives differ at most by multiplication by one global $f \in F^\times$. Thus their arithmetic line classes differ by $\text{div}(f)$, whose degree is zero by the product formula. Hence

$$(8) \quad \deg W_{\text{obj}} = \deg Q_{\text{obj}}.$$

Remark 8.3 (Why no false tensor-power strengthening follows). The argument uses the source-specific pilot map and the theta-function cyclotomic/Kummer synchronization for the original theta-pilot. The source explicitly warns that this input does not extend formally to arbitrary tensor powers. Thus the proof does not manufacture the known-to-be-false sharper inequalities for $T_\Theta^{\otimes N}$.

9. CONSTRUCTION OF AN ACTUAL POSSIBLE IMAGE

Let

$$W = \text{Reg}(W_{\text{obj}})$$

be the local fractional-ideal realization of W_{obj} in the target mono-analytic tensor packets. To use volume monotonicity, one must prove actual membership

$$W \subseteq \mathcal{U}_{\Theta},$$

not merely abstract isomorphism with a q -object.

Lemma 9.1 (Ordinary Kummer images are permitted by Ind3). *At every nonarchimedean place, the vertically coric tensor packet used in IUT III contains the image of the ordinary Kummer isomorphism. At every archimedean place, the corresponding closed ball also contains the ordinary Kummer image. Hence the path with no extra log-link iterate is an actual member of the Ind3 family.*

Proof. This is exactly the first class of images in IUT III, Proposition 3.5(ii)(a) at nonarchimedean places; the second class consists of Kummer maps precomposed with positive log-link iterates. Proposition 3.5(ii)(b) includes the ordinary Kummer image in the archimedean closed ball and separately records the subsets mapping onto images of iterates. $\square$

Lemma 9.2 (The selected single-link path is a genuine possible image). *The region W defined by (7) belongs to the theta-pilot possible-image union $\mathcal{U}_{\Theta}$.*

Proof. Choose the data as one synchronized tuple:

- (1) the horizontal representative is the fixed h_{00}^{MOD} of Theorem 7.6;
- (2) the Ind1 transport is the forced coric map $P_s^{\text{MOD}} = K_{10}^{\text{MOD}} h_{00}^{\text{MOD}} (K_{00}^{\text{MOD}})^{-1}$;
- (3) the Ind2 representative is the unit component of the same h_{00}^{MOD} ;
- (4) the Ind3 choice is the ordinary Kummer branch, including the compatible action of the split LGP monoid, with no positive log-link pre-iterate.

The first two choices are allowed by fullness and functoriality of processions. The third is coherent with the value component by the insulation property of Theorem 7.5. The fourth is allowed by Theorem 9.1. Proposition 3.7(v) assembles the resulting local fractional ideals into one global LGP-Frobenioid object. Thus the tuple is one of the choices over which the possible-image union is taken, and its realization is W .

If an isomorphic fractional-ideal representative is produced, the discrepancy is multiplication by one global rational function. The global number-field action is part of the multiradial representation, so this translate is again represented in the same possible-image relation; the product formula also leaves the global degree unchanged. $\square$

Remark 9.3 (Second principal expert-review point). The last sentence uses the source claim that the global number-field monoid action is part of the same possible-image relation. A complete machine formalization must encode this action and Proposition 3.7(v), rather than merely the abstract statement that isomorphic F_{mod} objects have equal degree.

10. DEGREE-VOLUME REFLECTION AND THE CORE INEQUALITY

Let μ_{proc} be the packet-normalized signed log-volume, averaged over procession capsules. We use two standard properties:

- (i) monotonicity: $A \subseteq B$ implies $\mu_{\text{proc}}(A) \leq \mu_{\text{proc}}(B)$;
- (ii) degree-volume compatibility: if a region is realized by a global F_{mod} object L , then

$$\mu_{\text{proc}}(\text{Reg } L) = \deg L.$$

The second property is the content of IUT III, Proposition 3.9(iii); the first is the monotonicity of Haar volume and its logarithm in the admissible finite-volume regime.

By (8),

$$\begin{aligned}\mu_{\text{proc}}(W) &= \deg W_{\text{obj}} \\ &= \deg Q_{\text{obj}} \\ &= -|\log q|.\end{aligned}$$

The final equality is the definition of the native q -pilot signed volume in Corollary 3.12. By Theorem 9.2, $W \subseteq \mathcal{U}_\Theta$, hence

$$-|\log q| = \mu_{\text{proc}}(W) \leq \mu_{\text{proc}}(\mathcal{U}_\Theta) = -|\log \Theta|.$$

This proves Theorem 4.1.

Corollary 10.1 (IUT III, Corollary 3.12). *If*

$$-|\log \Theta| \leq C_\Theta |\log q|, \quad |\log q| > 0,$$

then $C_\Theta \geq -1$.

Proof. The core inequality gives

$$-|\log q| \leq -|\log \Theta| \leq C_\Theta |\log q|.$$

Divide by the positive number $|\log q|$. □

11. FROM COROLLARY 3.12 TO THE abc INEQUALITY

IUT IV, Theorem 1.10 inserts Theorem 10.1 into explicit local estimates and obtains, in the notation of that paper,

$$(9) \quad \frac{1}{6} \log q \leq \left(1 + \varepsilon + \frac{28d_{\text{mod}}}{\ell}\right) (\log d_F + \log f_F) + 2 \log(\ell \varepsilon^{-7}).$$

The factor $1/6$ comes from the procession average of the Gaussian exponents j^2 .

After the general-position reduction, IUT IV states the following height inequality. Let X be a smooth, proper, geometrically connected curve over a number field, let D be a reduced divisor, and assume $X \setminus D$ is hyperbolic. For points of bounded degree,

$$(10) \quad h_{\omega_X(D)}(P) \leq (1 + \varepsilon) (\text{LogDiff}_X(P) + \text{LogCond}_D(P)) + O(1).$$

We now derive abc in ordinary language. Take

$$X = \mathbb{P}^1, \quad D = \{0, 1, \infty\}, \quad P = \frac{a}{c},$$

where $a + b = c$ and $\gcd(a, b, c) = 1$. Since

$$\omega_{\mathbb{P}^1}(D) \cong \mathcal{O}_{\mathbb{P}^1}(1),$$

the left side of (10) is

$$\log \max\{|a|, |b|, |c|\} + O(1).$$

Because P is rational, its logarithmic different term is zero. The reduction of P meets one of $0, 1, \infty$ modulo a prime p precisely when p divides a , b , or c . Therefore

$$\text{LogCond}_D(P) = \sum_{p|abc} \log p + O(1) = \log \text{rad}(abc) + O(1).$$

Substituting into (10) gives

$$\log \max\{|a|, |b|, |c|\} \leq (1 + \varepsilon) \log \text{rad}(abc) + C_\varepsilon,$$

which is (1). Exponentiating yields

$$\max\{|a|, |b|, |c|\} \leq K_\varepsilon \text{rad}(abc)^{1+\varepsilon}.$$

Theorem 11.1 (Candidate abc theorem). *Assume the IUT I–IV constructions and estimates cited above, and assume that the common-source and scale-rigidity instantiations in Theorems 7.3, 8.2 and 9.2 are valid. Then the abc conjecture holds.*

Proof. Theorems 7.3 and 8.2, together with Lemma 9.2, give the witness in Theorem 4.1. This gives Corollary 10.1; IUT IV gives (10); the specialization above gives (1). $\square$

12. INDEPENDENT ROUTES AND THEIR PRESENT STATUS

The following routes were investigated independently.

Route	Status	Decisive point
Common-source synchronization	IUT strongest candidate	One underlying prime-strip isomorphism generates all components; packet normalization fixes the realified scale.
Arithmetic Teichmüller Spaces	independent claimed closure	Requires a proved identification of the constructed theta locus and volume normalization with the IUT possible-image locus.
Analytic exceptional-set bounds	substantial partial result	A power-saving count permits an infinite sufficiently sparse exceptional sequence.
Fixed- S unit equations and Baker theory	insufficient uniformly	Existing height bounds depend too strongly on the size and arithmetic of S to yield exponent $1 + \varepsilon$ uniformly.
Frey curves and modularity	equivalent bottleneck	The missing uniform discriminant-conductor estimate is essentially Szpiro/ <i>abc</i> .
Mason-Stothers/arithmetic derivative	function-field success only	Every ordinary derivation from $\mathbb{Z}$ is zero; a global compatible family of nontrivial p -derivations with the required truncated counting inequality is absent.
Independent unit/value witnesses	ruled out	The anti-diagonal finite relation is a counterexample; a common source parameter or insulation theorem is necessary.
Approximation below	from not weaker	For a closed half-line of readings it is equivalent to the desired inequality.
Degree equality out membership	with-ruled out	Equal numerical readings do not imply that the object lies in the actual possible-image relation.

No non-IUT route presently supplies the missing uniform number-field theorem. Within the IUT route, the common-source plus scale-rigidity construction is the only direction not eliminated by a concrete obstruction.

13. COUNTEREXAMPLES DELIMITING THE ARGUMENT

The following elementary counterexamples explain why the proof must construct one coherent possible image rather than rely on weaker slogans.

Counterexample 13.1 (Separate unit and value witnesses do not assemble). Let $U = V = \{0, 1\}$ and

$$S = \{(1, 0), (0, 1)\} \subseteq U \times V.$$

The desired unit value 1 occurs in the first projection and the desired value-group value 1 occurs in the second projection, but $(1, 1) \notin S$. Projection-wise existence therefore does not imply a coherent output. The proof above uses a single structured horizontal representative.

Counterexample 13.2 (Compatibility alone does not imply the numerical bridge). Take two two-point sets, let the link be the identity, let both algorithms be Boolean negation, and let the input pilot be 0. The algorithm commutes with the link, but the output is $1 \neq 0$. Define $\text{vol}(0) = -1$ and $\text{vol}(1) = -2$. The link preserves volume, yet the desired inequality $-1 \leq -2$ fails. One therefore needs actual pilot realization, not merely parallel transport and volume preservation.

Proposition 13.3 (Approximation from below is not a weaker substitute). *If the possible readings form the closed half-line $(-\infty, u]$, then*

$$[\forall \delta > 0 \exists s \leq u : L - \delta \leq s] \iff L \leq u.$$

Proof. The forward implication follows by contradiction: if $L > u$, take $\delta = (L - u)/2$. The reverse implication is obtained by taking $s = L$. $\square$

14. LEAN FORMALIZATION AND EXACT VERIFICATION BOUNDARY

The accompanying Lean 4.32/Mathlib development contains the following machine-checked modules.

Module	Verified content
<code>CoarsifiedFullPoly.lean</code>	Set-level and category-level conjugate equivalence; membership in a full family; natural isomorphism $HK_1 \cong K_0P$; and objectwise evaluation.
<code>GlobalPrincipalReferenceImage.lean</code>	Principal discrepancy has degree zero; a realized output region in the theta hull has the native q -degree; monotonicity gives the bridge.
<code>HDDSHENumericalReflection.lean</code>	Common-expression numerical reflection with explicit degree–volume and hull hypotheses.
<code>ABCProofChain.lean</code>	The bridge and upper estimate imply $C_\Theta \geq -1$; exponentiation of the logarithmic abc bound.
<code>StructuredSourceBoundary.lean</code>	The exact source-facing certificate required from the concrete IUT objects, and the theorem that such a certificate closes the numerical bridge.
<code>SingleLinkWitness.lean</code>	The single-link-first conjugate transport; construction of one possible witness from fullness and the ordinary Ind3 branch; conversion to the structured bridge certificate.
<code>IndeterminacySynchronization.lean</code>	Ind2 insulation: a group acting only on units leaves the value component fixed and cannot realize the anti-diagonal obstruction.
<code>CommonSourceSynchronization.lean</code>	Selection of one common source parameter and extraction of one coherent possible-image witness.
<code>PacketNormalizationRigidity.lean</code>	A comparison preserving the normalized response $-\log p$ has positive-real scale exactly one.

No theorem in those modules uses `sorry` or introduces a new axiom. The formalized theorems are conditional on explicit arguments and structures rather than hidden assumptions.

The following objects are *not* yet fully encoded:

- (1) the complete definitions of Hodge theaters and structured prime-strips;
- (2) the concrete full poly-isomorphism of the LGP-link;
- (3) the construction in Lean of the common source parameter set of IUT II, Corollary 4.10 and all of its functorial images in IUT III;
- (4) the expansion of Theorem 3.11(iii)(a)–(d) into one pointwise synchronized structured diagram, together with a proof that the local normalized response rigidifies the $\mathbb{R}_{>0}$ orbit exactly as in Theorem 8.2;
- (5) the local Ind3 inclusions and their global assembly in Proposition 3.7(v);
- (6) the degree–volume theorem for the actual Frobenioid implementation;
- (7) IUT IV’s general-position reduction and the arithmetic-geometric identification of (10) with the abc conductor.

Accordingly, a successful Lean build certifies the abstract proof corridor

$$\text{structured witness} \implies -|\log q| \leq -|\log \Theta| \implies C_\Theta \geq -1,$$

not the truth of every source-facing premise. This distinction is essential: a proof assistant verifies the theorem that has been formalized, not the intended interpretation of informal source terminology.

15. CONCLUSION

The candidate closure developed here has five synchronized steps:

- (1) choose one underlying source isomorphism and derive all horizontal, unit, value-group, number-field, localization, and procession maps functorially;
- (2) use the theta-function and number-field insulation results to prevent componentwise anti-diagonal assembly;
- (3) use packet-normalized response to force the positive-real degree scale to be one;
- (4) choose the ordinary Kummer branch and globalize it to one actual possible-image region W ;
- (5) apply degree-volume compatibility and monotonicity.

This gives

$$\mu_{\text{proc}}(W) = -|\log q| \leq \mu_{\text{proc}}(\mathcal{U}_{\Theta}) = -|\log \Theta|,$$

and the remaining IUT IV chain yields the standard *abc* inequality.

The ordinary mathematical argument is now closed once the source statements are read pointwise and functorially in the sense of [Theorems 7.3](#) and [8.2](#). What is not closed by the present work is the *independent certification of that source reading*: neither this paper nor the current public Lean libraries encode enough of the actual Hodge-theater and Frobenioid structures to make Lean construct the common-source scale-one certificate from first principles.

Accordingly, the strongest justified status is:

a finite, source-grounded proof candidate with a precise formalization boundary.

If specialists verify the common-source and scale-rigidity instantiation, or if a full Lean development constructs it, the displayed chain constitutes a proof of the *abc* conjecture. Until then it should not be advertised as an independently certified resolution.

APPENDIX A. DEPENDENCY GRAPH

The complete claimed chain is

$$\begin{array}{ccccc} \text{IUT I-II constructions} & \longrightarrow & \text{IUT III, Theorem 3.11} & \xrightarrow{\text{common source} + \text{scale rigidity} + \text{possible image}} & -|\log q| \leq -|\log \Theta| \\ & & & & \downarrow \\ & & & & C_{\Theta} \geq -1 \\ & \longleftarrow & \text{IUT IV height inequality} & \longleftarrow & \end{array}$$

The diagonal arrow supplied by this paper is exactly the source-facing bridge from [Theorem 3.11](#) to the numerical inequality.

APPENDIX B. REPRODUCIBILITY CHECKLIST

For independent review, the following should be checked in order.

1. Verify the domains and codomains of K_{00}^{MOD} , K_{10}^{MOD} , and P_{01}^{MOD} at the full structured prime-strip level.
2. Verify that the horizontal LGP-link is full for exactly those structured objects.
3. Choose one horizontal representative first, form its forced conjugate transport, and verify that all four components of [Theorem 3.11\(iii\)](#) are induced by that same representative.
4. Evaluate the resulting natural isomorphism at the theta-pilot and check that every representative is compatible with the pilot's generator-up-to-torsion definition.
5. Apply $F_{\text{MOD}} \simeq F_{\text{mod}}$ only in the target column.

6. Check that the ordinary Kummer image is included locally at every place and that Proposition 3.7(v) globalizes the same family.
7. Check degree–volume normalization and the sign conventions.
8. Verify the downstream constants in IUT IV and the specialization to $\mathbb{P}^1 \setminus \{0, 1, \infty\}$.

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